

PAC-NeRF: P2G $\leftrightarrow$ G2P

NeRF uses Eulerian Field

Eulerian Grid

$$\mathcal{F}_p^P \approx \sum_i w_{ip} \mathcal{F}_i^G$$

$p \sim$ particles

$w \sim$ weights

Used for rendering

MPM uses Lagrangian Field

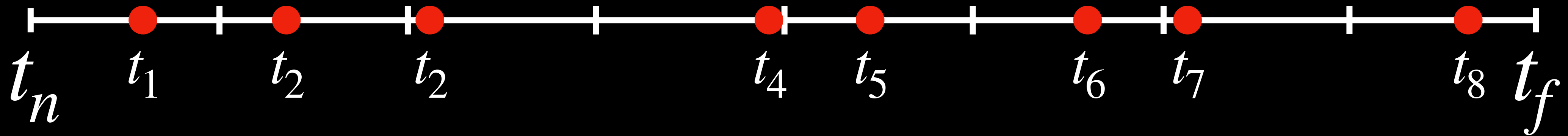
Lagrangian Grid

$$\mathcal{F}_i^G \approx \frac{\sum_p w_{ip} \mathcal{F}_p^P}{\sum_p w_{ip}}$$

Barycentric Interpolation

Used for advection process

PAC-NeRF: Volume Rendering Continuum



Pixel Appearance:

$$\mathbf{C}(\mathbf{r}, t) = \int_{s_n}^{s_f} T(s, t) \sigma(\mathbf{r}(s), t) \mathbf{c}(\mathbf{r}(s), \omega, t) ds + \mathbf{c}_{bg} T(s_f, t)$$
$$T(s, t) = \exp \left(- \int_{s_n}^s \sigma(\mathbf{r}(\bar{s}), t) d\bar{s} \right)$$